

UNIT IV- Orthogonalization, Eigenvalues and Eigenvectors:

Question and Answers

1) The vectors $q_1 = (1, 0, 0)$, $q_2 = (0, 3, 4)$ and $q_3 = (0, 4, -3/5)$ form an orthonormal basis for R^3 . Express the vector $v = (7, -5, 10)$ as a linear combination of the q 's.

$$\text{Solution: } \begin{pmatrix} 7 \\ -5 \\ 10 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 4 \\ -3 \end{pmatrix}$$
$$7 = c_1 + 0 + 0.$$

Solving the equations $-5 = 0c_1 + 3c_2 + 4c_3$, we get $c_1 = 7, c_2 = \frac{5}{3}, c_3 = -\frac{15}{4}$
 $10 = 0c_1 + 3c_2 - 4c_3$

$$\text{Therefore } v = 7q_1 + \frac{5}{3}q_2 - \frac{15}{4}q_3$$

2) Find a third column so that the matrix $Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{14}} & - & - & - \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{14}} & - & - & - \\ \frac{1}{\sqrt{3}} & \frac{-3}{\sqrt{14}} & - & - & - \end{bmatrix}$ is orthogonal.

$$\text{Solution: Let } Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{14}} & x \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{14}} & y \\ \frac{1}{\sqrt{3}} & \frac{-3}{\sqrt{14}} & z \end{bmatrix} \text{ i. e, third columns elements are } (x, y, z)$$

Since Q has be orthogonal then

$$a^T c = 0 \Rightarrow \left(\frac{1}{\sqrt{3}} \frac{1}{\sqrt{3}} \frac{1}{\sqrt{3}} \right) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow x + y + z = 0 \dots (1)$$

$$b^T c = 0 \Rightarrow \left(\frac{1}{\sqrt{14}} \frac{2}{\sqrt{14}} \frac{-3}{\sqrt{14}} \right) \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow x + 2y - 3z = 0 \dots (2)$$

Solving equation (1) and equation (2), by taking $z = 1$ (because z is the free variable) we get, $x = -5$ and $y = 4$.

$$\text{Therefore } c = \begin{pmatrix} -5 \\ 4 \\ 1 \end{pmatrix} \text{ OR } c = \begin{pmatrix} 5 \\ -4 \\ -1 \end{pmatrix}.$$

$$\text{Let } q_3 = \pm \begin{pmatrix} \frac{-5}{\sqrt{42}} \\ \frac{4}{\sqrt{42}} \\ \frac{1}{\sqrt{42}} \end{pmatrix} \text{ (After Normalization i.e., } \sqrt{(-5)^2 + (4)^2 + 1^2} = \sqrt{42} \text{)}$$

$$\text{Hence } Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{14}} & \frac{-5}{\sqrt{42}} \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{14}} & \frac{4}{\sqrt{42}} \\ \frac{1}{\sqrt{3}} & \frac{-3}{\sqrt{14}} & \frac{1}{\sqrt{42}} \end{bmatrix} \text{ OR } Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{14}} & \frac{+5}{\sqrt{42}} \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{14}} & \frac{-4}{\sqrt{42}} \\ \frac{1}{\sqrt{3}} & \frac{-3}{\sqrt{14}} & \frac{-1}{\sqrt{42}} \end{bmatrix}$$

3) Use the Gram – Schmidt process to find a set of orthonormal vectors from the independent vectors $a_1 = (1, 1, 1)$, $a_2 = (0, 1, 1)$ and $a_3 = (0, 0, 1)$. Also find the $A = QR$ factorization where $A = [a_1 \ a_2 \ a_3]$.

$$\text{Solution: Let } a = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, b = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \text{ and } c = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$q_1 = \frac{a}{\|a\|} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$e = b - (q_1^T b) q_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \left(\begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right) \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix}$$

$$q_2 = \frac{e}{\|e\|}, \|e\| = \sqrt{\left(\frac{-2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2} = \frac{\sqrt{6}}{3}$$

$$q_2 = \frac{e}{\|e\|} = \frac{3}{\sqrt{6}} \begin{pmatrix} \frac{-2}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} = \begin{pmatrix} \frac{-2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix}$$

$$\begin{aligned}
 e_1 &= c - (q_1^T c)q_1 - (q_2^T c)q_2 \\
 &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} - \left(\begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \left(\begin{bmatrix} \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) \begin{pmatrix} \frac{-2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix} \\
 &= \begin{pmatrix} 0 \\ \frac{-1}{2} \\ \frac{1}{2} \end{pmatrix}
 \end{aligned}$$

$$q_3 = \frac{e_1}{\|e_1\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \frac{-1}{2} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \quad \|e_1\| = \sqrt{(0)^2 + \left(\frac{-1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{1}{\sqrt{2}}$$

Therefore $Q = \begin{pmatrix} q_1 & q_2 & q_3 \\ \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{pmatrix}$

TO find R:

We have $A=QR$,

$$R = Q^T A = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \sqrt{3} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$OR R = \begin{pmatrix} q_1^T a & q_1^T b & q_1^T c \\ 0 & q_2^T b & q_2^T c \\ 0 & 0 & q_3^T c \end{pmatrix} = \begin{pmatrix} \sqrt{3} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$

4) Find the eigenvalues of A , A^2 , A^{-1} and $A + 4I$ if

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Solution: Consider the characteristic equation $|A - \lambda I| = 0$

$$\text{i.e., } |A - \lambda I| = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (2 - \lambda)(2 - \lambda) + 2 = 0$$

$$\lambda = 1, 3$$

Eigen values of A are 1, 3.

We know that the eigen values of A^{-1} is $\frac{1}{\lambda}$, if eigen values of A is λ .

Hence the eigen values of $A^{-1} = 1, \frac{1}{3}$

The eigen values of A^2 is $1^2, 3^2 = 1, 9$.

Eigen values of $A + 4I$: Here we need to shift λ to $\lambda + 4$

Eigen value of $A + 4I$: $1 + 4 = 5, 3 + 4 = 7$.

5) Factor $A = \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$ and hence compute A^{100} .

Solution: To find the eigen values of A :

Consider the characteristic equation $|A - \lambda I| = 0$

$$\text{i.e., } |A - \lambda I| = \begin{vmatrix} 4 - \lambda & -3 \\ -1 & 2 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (4 - \lambda)(2 - \lambda) - 3 = 0$$

$$\lambda^2 - 6\lambda + 5 = 0$$

$$\lambda = 1, 5$$

Consider $(A - \lambda I)X = 0$

$$\begin{bmatrix} 4 - \lambda & -3 \\ -1 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Case 1: When $\lambda = 1$.

$$\begin{bmatrix} 3 & -3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$3x - 3y = 0, -x + y = 0 \Rightarrow x = y$$

Since both equations are same $2(\text{unknowns}) - 1(\text{equation}) = 1$ free variable. Let $y = 1$ be the free variable, then $x = 1$,

Therefore, the eigen vector is

$$X_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Case 2: When $\lambda = 5$.

$$\begin{bmatrix} -1 & -3 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x - 3y = 0, -x - 3y = 0 \Rightarrow x = -3y$$

Since both equations are same $2(\text{unknowns}) - 1(\text{equation}) = 1$ free variable. Let $y = 1$ be the free variable, then $x = -3$,

Therefore, the eigen vector is

$$X_2 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$\text{Therefore } S = [X_2, X_1] = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}, \quad S^{-1} = \frac{1}{-4} \begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix}$$

We have $A = S \wedge S^{-1}$ and $A^{100} = S \wedge^{100} S^{-1}$

$$\begin{aligned}
 &= \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}^{100} - \frac{1}{4} \begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix} \\
 &= -\frac{1}{4} \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 5^{100} & 0 \\ 0 & 1^{100} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix}
 \end{aligned}$$